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$\gamma\delta$ T-cell autoresponses to ectopic membrane proteins: a new type of pattern recognition

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T-cell receptor (TCR) $\gamma\delta$ -expressing cells are conserved lymphocytes of innate immunity involved in first-line defense and immune surveillance. TCR $\gamma\delta$ recognizes protein/nonprotein ligands without the help of the major histocompatibility complex (MHC), especially via direct binding to protein ligands, which is dependent primarily on the δ chain complementary determining region 3 (CDR3 δ). However, the mechanism of protein–antigen recognition by human $\gamma\delta$ TCRs remains poorly defined. We hypothesize that $\gamma\delta$ TCRs recognize self-proteins expressed ectopically on the cell membrane that are derived from intracellular components under stress. Here, we mapped 16 intercellular self-proteins among 21,000 proteins with a huProteinChip as putative ligands for V δ 1/V δ 2 TCRs, 13 for V δ 1 TCRs and 3 for V δ 2 TCRs. Functional tests confirmed that ectopic nucleolin (NCL) is a ligand for the V δ 1 TCR, whereas protein-glutamine γ -glutamyltransferase K (TGM1) is a ligand for the V δ 2 TCR. In the context of radiation exposure, the ectopic expression of intracellular proteins on the tumor cell surface is related to the increased antitumor cytotoxicity of $\gamma\delta$ T cells both in vitro and in vivo. In conclusion, the recognition of intracellular proteins that are ectopically expressed on somatic cells by human $\gamma\delta$ TCRs is a basic interaction mechanism that enables new types of immune pattern recognition and a novel $\gamma\delta$ TCR-ligand-based strategy for tumor immunotherapy.

Keywords: $\gamma\delta$ T-cell; Ectopic proteins; Pattern recognition; Stress condition; Immunotherapy

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INTRODUCTION

In contrast to conventional $\alpha\beta$ T cells, human $\gamma\delta$ T cells possess several characteristics of innate immune cells [1–7], such as limited diversity and specificity of $\gamma\delta$ T-cell receptors (TCRs) for only certain antigens, without major histocompatibility complex (MHC) restriction [8, 9]. In recent decades, research on the mechanism of antigen recognition by $\gamma\delta$ TCRs has focused on MHC-related molecules, as well as foreign lipids, glycans and phosphates [4, 10–14]. However, few $\gamma\delta$ TCRs specific for protein antigens, which are more diverse and stable in structure than nonpeptidic antigens are well characterized. Previously, we reported CDR3 δ -dependent direct binding of the $\gamma\delta$ TCR to its protein ligand [1, 15, 16] and identified human mutS homolog 2 (hMSH2) as the ligand for the V δ 2 TCR. hMSH2 is a DNA mismatch repair protein expressed in the nucleolus and ectopically expressed on the cell membrane under stress conditions; this protein is incidentally captured by the CDR3 δ peptide designed by our group [17]. We also found that human $\gamma\delta$ TCRs primarily use conserved flanking regions of both the CDR3 δ 2 and CDR3 δ 1 sequences to bind ligands, which are limited in number and share some of the same motifs in the gene sequence [16, 18]. Since then, we hypothesized that stress-induced ectopic expression of intracellular self-proteins might generate a potential pool of self-antigens for the $\gamma\delta$ TCR repertoire. In other words,

$\gamma\delta$ TCRs recognize intracellular self-proteins on the somatic cell membrane as nonself antigens under stress. In this simple way, $\gamma\delta$ TCRs can sense damage to somatic cells and rapidly respond to a series of factors, including cytotoxicity and cytokine production, possibly resulting in the recognition of an innate $\gamma\delta$ T-cell-mediated damage-associated molecular pattern (DAMP). Here, we report our findings to confirm this hypothesis.

RESULTS

CDR3 δ can specifically bind to tumors

Our previous studies demonstrated that the antigen specificity of human $\gamma\delta$ TCRs primarily depends on the conserved region in CDR3 δ [16, 18]. Two dominant CDR3 δ sequences have been identified: a GTM from a V δ 1 TCR expressed by gastric tumor-infiltrating $\gamma\delta$ T cells [16] and OT3 from a V δ 2 TCR expressed by epithelial ovarian cancer-infiltrating $\gamma\delta$ T cells [15]. Two types of $\gamma\delta$ TCR-expressing cells based on these two CDR3 δ sequences exhibited strong antitumor activities both in vitro and in vivo [19, 20].

In the present study, we synthesized two biotin-labeled CDR3 δ peptides, GTM and OT3, and performed immunohistochemistry (IHC) with MC246a tumor tissue chips containing 12 different

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organ tumor tissues and matched paracarcinoma tissues. The results showed that both the GTM (Fig. 1a) and OT3 (Fig. 1b) peptides could bind to various different tumor tissues compared with the paracarcinoma controls. Flow cytometric analysis of 50 human tumor cell lines (Supplementary Data Table 2) revealed that although the binding patterns were quite different, both peptides bound to various tumor cell lines but showed low binding activities to human PBMCs, which served as a normal control (Fig. 1c, d). Human $\gamma\delta$ T cells can directly recognize stress-induced proteins, such as the UL16 binding protein (ULBP) family [21], MHC I-related A/B (MICA/B) [22], MSH2 [17], and heat shock protein (HSP) family proteins [23], through both the $\gamma\delta$ TCR and the natural killer (NK) cell receptor NKG2D. An MST revealed that the GTM and OT3 peptides could also bind to some of these known ligands (Fig. 1e, f).

Next, we designed two CAR molecules with single-chain V γ 9-V δ 2 (OT3) and V γ 4-V δ 1 (GTM) as their extracellular domains, which replaced the conventional single-chain Fv fragment (Fig. 1g), to evaluate whether the structure can specifically bind to tumor cells and mediate CAR-T-cell cytotoxicity. The $\gamma\delta$ CAR molecule was inserted into the lentiviral vector cFUGW, which contains the CD28 transmembrane domain, and 4-1BB and CD3 ζ , which are classic CAR molecules. We packaged the viruses and infected q β T cells. The expression of exogenous $\gamma\delta$ CAR molecules was validated by Western blotting and flow cytometry of the GFP signal (Supplementary Data Fig. 1). When coincubated with tumor cells, $\gamma\delta$ CAR-T cells presented increased expression levels of CD69, TNF- α (Fig. 1h) and granzyme B (Fig. 1i). LDH release assays were used to measure the cytotoxicity of $\gamma\delta$ CAR-T cells against different tumor cell lines in vitro. OT3 $\gamma\delta$ CAR-T cells exhibited stronger cytotoxicity against OVCAR-8, Raji and NCI-H2228 cells (Fig. 1j), and GTM $\gamma\delta$ CAR-T cells exhibited stronger cytotoxicity against HR8348, LoVo, OVCAR-8 and SKOV3 cells (Fig. 1k).

To further confirm the specific targeting of $\gamma\delta$ CAR to tumor-related proteins, NOD-Prkdc^{em26cas9d52}Il2rg^{em26cas9d22}Nju (NCG) mice transplanted with OVCAR-8 cells were used as a model. Compared with control T cells, OT3 $\gamma\delta$ CAR-T cells exhibited stronger cytotoxic activity against tumor cells in vivo, as tumor growth was significantly inhibited (Fig. 1l, m), and the survival time of the tumor-bearing mice was significantly prolonged (Fig. 1n). Taken together, our results suggest the specific binding and functions of CDR3 δ 1 GTM and CDR3 δ 2 OT3 to tumor-related antigens, which can serve as probes to screen the protein ligands of human $\gamma\delta$ TCRs.

Mapping protein ligands for $\gamma\delta$ TCRs

On the basis of two tumor-specific CDR3 δ sequences derived from $\gamma\delta$ TILs, we then applied a recombinant GTM-grafted V γ 4 δ 1-Fc fusion protein and GTM-grafted soluble $\gamma\delta$ TCR as probes to identify the potential protein ligands for the V δ 1 TCR, and we used a biotin-OT3 peptide to identify the potential protein ligands of the V δ 2 TCR; these ligands were identified through high-throughput screening via a HuProt™ human proteome microarray, which provides the largest number of unique human proteins known to be included on a single slide, allowing thousands of interactions to be profiled in a high-throughput manner [24] (Fig. 2a). Among the 21,000 proteins, 13 proteins were found to specifically bind to the GTM-grafted V γ 4 δ 1-Fc fusion protein and GTM-grafted soluble $\gamma\delta$ TCR (Fig. 2b, c), whereas 3 proteins were found to bind to the biotin-labeled OT3 peptide (Fig. 2d, Supplementary Data Table 3). These results suggest that $\gamma\delta$ TCRs recognize a limited variety of protein ligands in a pattern recognition-like manner. Impressively, all of these identified potential protein ligands are physiologically located in an intercellular compartment—either the nucleus or the cytosol.

Ectopic NCL, a ligand for the V δ 1 TCR

Next, further studies were performed to functionally verify the protein ligands of $\gamma\delta$ T cells. We validated hMSH2, a protein that exhibits stress-induced ectopic expression, as a new functional

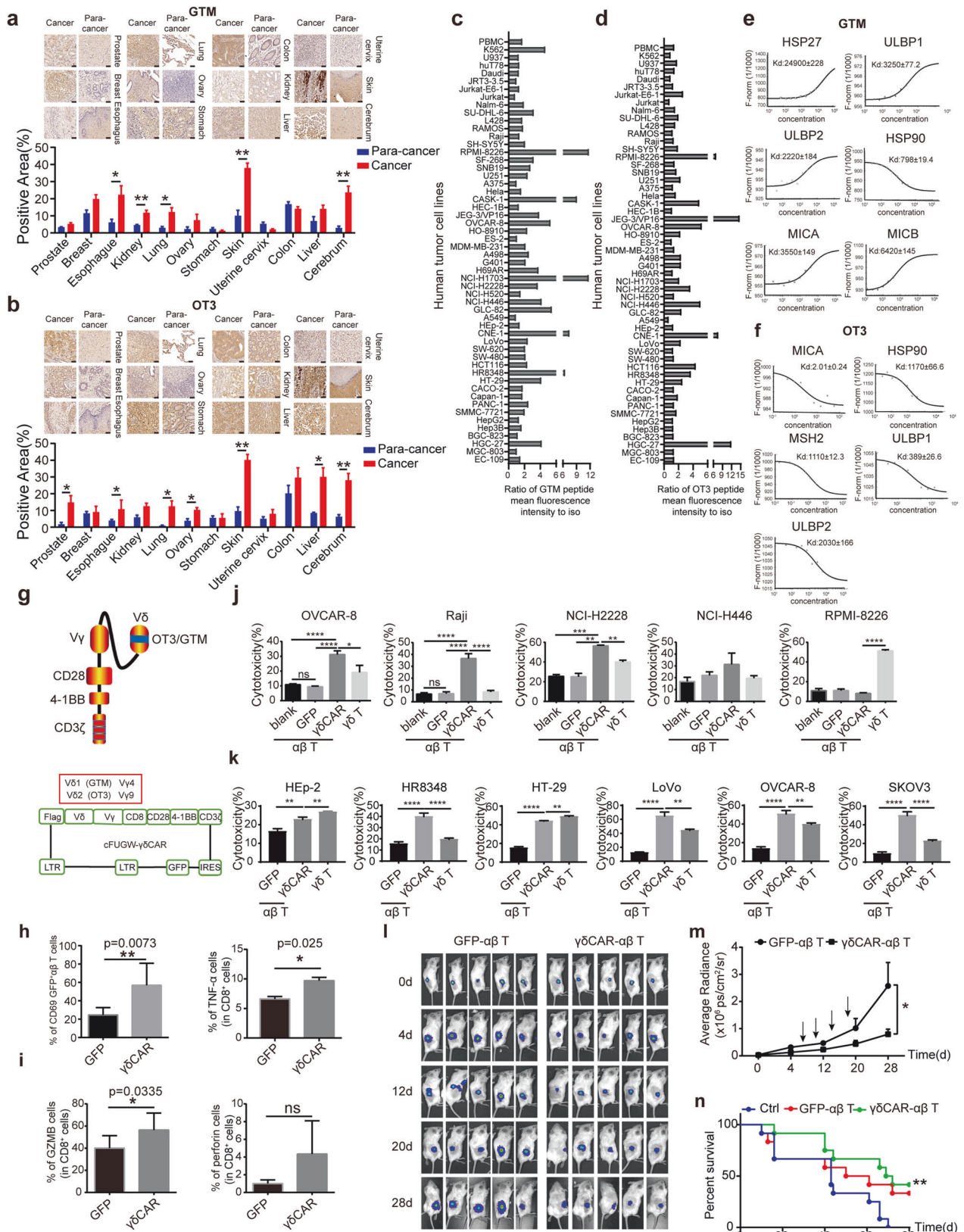
protein ligand for the V δ 2 TCR. hMSH2 is recognized by $\gamma\delta$ T cells to trigger cytotoxicity against tumor cells [17]. Thus, we selected NCL, a candidate ligand for the V δ 1 TCR, to validate whether this new recognition mechanism is suitable for all the major human $\gamma\delta$ T-cell subsets, both V δ 1 and V δ 2 T cells.

NCL, a highly conserved RNA-binding protein with multiple functions in several cellular processes, such as chromatin remodeling, rRNA synthesis, mRNA processing, ribosome assembly, and nucleocytoplasmic transport [25], is expressed mainly in the nucleus. However, under conditions of cellular stress or in tumor cells, NCL is upregulated and translocated to the cytosol and cell surface membrane to display danger signals and induce antitumor responses [26].

Although the direct binding of GTM, GTM-Fc or γ 4 δ 1(GTM)-TCR to the recombinant NCL protein was not detected by the MST assay, which may be due to the lack of necessary posttranslational modifications or chaperones of the recombinant NCL protein in vitro, the interaction between GTM and the NCL-Fc fusion protein was confirmed by an immunoprecipitation assay (Supplementary Data Fig. 2). Immunohistochemistry assays with MC246a tumor tissue chips were conducted to examine the levels of NCL expression in tumor tissues, and the results revealed that NCL expression was upregulated in a variety of human tumor tissues compared with their paracarcinoma control tissues (Fig. 3a). The immunohistochemical results also revealed high expression of NCL in tumor biopsy samples from 4 of the 6 patients with primary hepatocellular carcinoma (Fig. 3b, Supplementary Data Fig. 3). Flow cytometry was also performed to detect the ectopic expression of NCL on the surface of 17 different human tumor cell lines. Despite the variations among the tumor cell lines, some exhibited high levels of ectopic NCL expression on the cell surface (Fig. 3c). We also performed immunofluorescence assays to examine both the ectopic expression and the nuclear expression of NCL in BGC-823 cells and HT-29 cells (Fig. 3d). The results revealed that most of the NCL proteins were in the nucleus, but some were ectopically expressed on the cell surface.

In addition to tumors, to determine whether the ectopic expression of NCL results from cell stress, we established a DNA damage-induced cell stress model with menadione [8] (Supplementary Data Fig. 4a–4c), which is widely used as an oxidative injury-inducing agent that causes reduced cell viability, increased poly ADP-ribose (PAR) accumulation, and increased sensitivity to cytotoxic effects [7, 22]. Menadione treatment caused a significant increase in the expression of stress-inducible $\gamma\delta$ TCR ligands, including ULBP1, ULBP4, hMSH2 and HSP70 (Supplementary Data Fig. 4d), as well as NCL (Fig. 3e), in 293 T cells. In tumor cell lines, menadione treatment-induced stress also significantly upregulated the expression of NCL (Fig. 3f) and hMSH2 (Supplementary Data Fig. 4e) on the cell surface. Collectively, these findings suggest that NCL is a stress-inducible molecule that is ectopically expressed in a variety of tumor cells or normal cells under stress conditions.

To further verify whether NCL is a ligand of $\gamma\delta$ T cells, we purified NCL proteins to amplify human $\gamma\delta$ T cells from PBMCs in vitro. The results showed that the recombinant NCL protein could specifically stimulate the proliferation of $\gamma\delta$ T cells, resulting in a significant increase in the $\gamma\delta$ T-cell percentage (Fig. 3g). In particular, we found that NCL predominantly stimulated the proliferation of V δ 1 T cells rather than V δ 2 T cells (Fig. 3h). The expression of the activation markers CD69 and CD25 on $\gamma\delta$ T cells was also significantly increased (Fig. 3i). Repeated experiments using sorted $\gamma\delta$ T cells (Supplementary Data Fig. 5) but not PBMCs yielded similar results (Supplementary Data Fig. 6). An in vitro V δ 1 T-cell expansion method was established for V δ 1 T cells with purities above 80% (Supplementary Data Fig. 7). Although there was no significant increase in the secretion of IFN- γ or IL-2 by V δ 1 T cells after NCL stimulation (Supplementary Data Fig. 8), blocking NCL with anti-NCL antibodies significantly decreased the cytotoxicity of V δ 1 T cells to MDA-MB-231 and SKOV3 cells (Fig. 3j, k).



Taken together, these findings confirmed that NCL is ectopically expressed on tumor cells or normal cells under conditions of stress and that NCL is a stress-inducible protein ligand of Vδ1 T cells. In addition to NCL, FBXO2, which is highly expressed in various

human colorectal cancer cell lines, also specifically binds to the GTM-grafted Vγδ1-Fc fusion protein in BLI (Supplementary Data Fig. 9), suggesting a potential ectopic protein ligand for the Vδ1 TCR for which further validation is needed.

Fig. 1 Two CDR3 δ peptides, GTM and OT3, with tumor binding specificity served as probes to screen the protein ligands of human $\gamma\delta$ TCRs. **a** and **b**. Immunohistochemical staining of a tumor tissue array with the biotin-labeled GTM peptide **a** and biotin-labeled OT3 peptide **b** followed by HRP-conjugated streptavidin. The binding was visualized using DAB as the substrate (brown) (400 $\times$). Cancer: tumor tissue; paracancerous: paracarcinoma tissues. Scale bar, 50 μ m. GTM and OT3 can specifically bind to several tumor cell lines. Fifty-three types of human tumor cell lines were stained with biotinylated (bio)-GTM **c** or (bio)-OT3 peptide **d** and APC-conjugated streptavidin. The binding activity was measured by flow cytometry. The flow data were standardized by the ratio between the mean fluorescence intensity of the peptide conjugate and the secondary antibody alone. The binding strength of the synthetic peptide was compared with that of normal human peripheral blood mononuclear cells (PBMCs). MST analysis of the binding activities of the GTM peptide **e** and OT3 peptide **f** to known $\gamma\delta$ TCR-recognized stress-inducible ligands. The K_d value for each protein was calculated. **g** Schematic diagram of OT3- and GTM- $\gamma\delta$ CAR molecules. A single-chain $V\gamma 9$ - $V\delta 2$ or $V\gamma 4$ - $V\delta 1$ structure, which is composed of a $V\delta 2$ TCR with the tumor-specific CDR3 δ sequence OT3 or $V\delta 1$ TCR with the tumor-specific CDR3 δ sequence GTM, forms the extracellular region of the $\gamma\delta$ CAR. The transmembrane and intracellular segments of a classical third-generation CAR (CD8-CD28-4-1BB-CD3 ζ) were linked to the $V\gamma$ - $V\delta$ sequence. $\gamma\delta$ CARs were cloned and inserted into the lentiviral vector cFUGW. Flow cytometry was used to assess the expression of the activation marker CD69, the secretion of TNF- α **h** and the cytotoxicity-related molecules granzyme B and perforin **i** in $\gamma\delta$ CAR-T cells after coculture with OVCAR-8 cells in vitro. The statistical data are representative of three independent experiments and are expressed as the means $\pm$ SDs. *, $p < 0.05$; **, $p < 0.01$. Cytotoxicity of OT3 CAR-T cells **j** and GTM CAR-T cells **k** in different tumor cells in vitro. Cytotoxicity was analyzed with an LDH cytotoxicity detection kit. The data are representative of three independent experiments and are expressed as the means $\pm$ SDs. OT3 CAR- $\alpha\beta$ T cells exhibit strong cytotoxic activity against tumors in an NSG mouse model. OVCAR-8-transplanted NSG mice were treated with GFP- $\alpha\beta$ T cells or OT3 CAR- $\alpha\beta$ T cells. Tumor growth in every mouse at each time point was measured by assessing the radiance of the tumor cells **l**. The tumor growth curve **m** and survival curve **n** of each group are shown

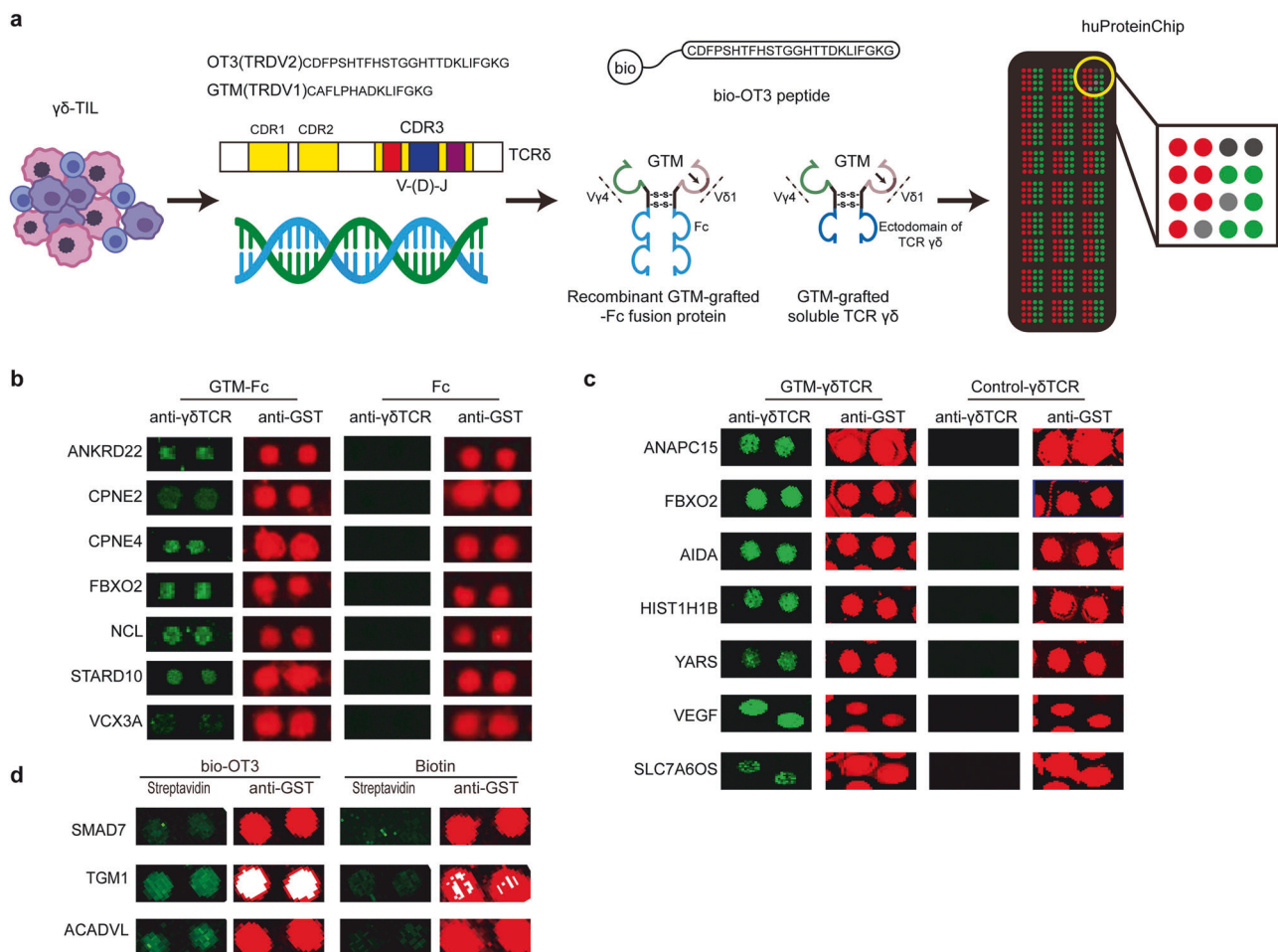
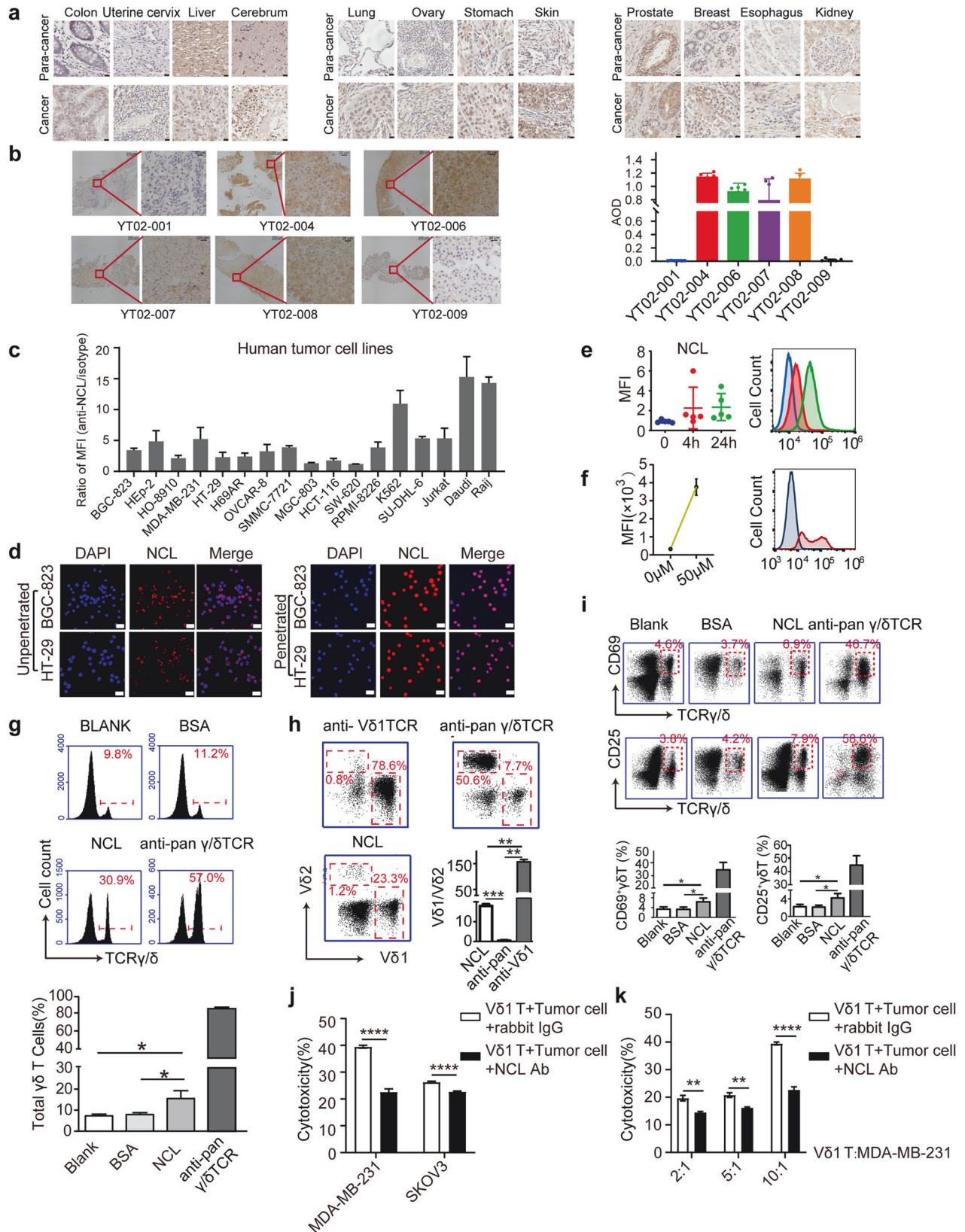


Fig. 2 Screening the potential protein ligands of $\gamma\delta$ T cells via a tumor-specific CDR3 δ sequence strategy. **a** Schematic diagram of the tumor-specific CDR3 δ sequence strategy. Tumor-infiltrating $\gamma\delta$ T cells were cultured, and their CDR3 δ sequences were analyzed. Two biotin-CDR3 δ peptides, GTM and OT3, were synthesized according to the dominant sequences, and their tumor binding specificities were verified. A recombinant GTM-grafted $V\gamma 4\delta 1$ -Fc fusion protein/GTM-grafted soluble $\gamma\delta$ TCR and biotin-OT3 peptide were subsequently used as probes to screen novel ligands of $\gamma\delta$ T cells via the huProteinChip. **b-d**. Screening the potential protein ligands of $\gamma\delta$ T cells via recombinant GTM-grafted $V\gamma 4\delta 1$ -Fc fusion protein **b** GTM-grafted soluble $\gamma\delta$ TCR **c** and biotin-labeled OT3 peptide **d** with the huProteinChip. The green fluorescence represents the specific binding signals of the probes, and the red fluorescence represents the protein loading control detected by anti-GST

Antitumor strategies targeting ectopic stress-inducing ligands

For the ectopic expression of NCL on the tumor cell membrane, we designed two types of genetic T cells on the basis of specific

antibodies targeting NCL to validate their antitumor efficacy: classic CAR-T cells, which had the CD28, 4-1BB and CD3 domains, and Ab- $\gamma\delta$ TCR-T cells, in which the VL and VH fragments of the



anti-NCL or anti-hMSH2 antibodies were used to replace the V fragment of the γ and δ chains, respectively, in the $\gamma\delta$ TCR (Fig. 4a). CAR-T cells and Ab- $\gamma\delta$ TCR-T cells targeting hMSH2 were used as positive controls. The expression of the Ab- $\gamma\delta$ TCR molecule and

CAR molecule in 293T cells was validated by Western blotting (Fig. 4b). After viral transduction, exogenous CAR⁺ and Ab- $\gamma\delta$ TCR⁺ $\alpha\beta$ T cells were analyzed by evaluating the ZsGreen signal via fluorescence microscopy (Supplementary Data Fig. 10) and

Fig. 3 Ectopically expressed NCL serves as a stress-induced ligand of V δ 1 T cells. **a** Immunohistochemical staining of a human tumor tissue microarray with anti-NCL antibodies. Cancer: tumor tissues; paracarcinoma: paracarcinoma tissues. Scale bar, 20 μ m. **b** Immunohistochemical staining of NCL in tumor biopsy samples from six HCC patients. The average optical density (AOD) value was measured to represent the expression level of NCL. Six fields of view were selected for each sample. The statistical data for each sample are shown on the right. Scale bar, 20 μ m. **c** Flow cytometric analysis of ectopic NCL protein expression on the surface of different human tumor cell lines. PBMCs served as the control. **d** Immunofluorescence assay examining the expression of NCL proteins on the cell surface and in the intracellular compartments of gastric adenocarcinoma BGC-823 cells and colorectal cancer HT-29 cells. Scale bar, 20 μ m. **e** Ectopic expression of NCL in menadione-treated 293 T cells was upregulated in a time-dependent manner. Three independent experiments were performed for statistical analysis (left), and representative results (right) are shown. **f** Short-term high-dose menadione treatment also upregulated the ectopic expression of NCL in 293 T cells. Three independent experiments were conducted for statistical analysis (left), and representative results are shown (right). **g** Flow cytometric analysis of the proliferation **g**, subtype **h** and activation marker expression **i** of $\gamma\delta$ T cells isolated from PBMCs and stimulated with the recombinant NCL protein. Three independent experiments were performed for statistical analysis, and representative results are shown. The anti-NCL antibody partially blocked the cytotoxic effects of V δ 1 T cells on different tumor cells **j** and different effect–target ratios **k**. Cytotoxicity was examined via an LDH assay, and V δ 1 T cells expanded with the anti-V δ 1 TCR were used as effectors. For all the flow cytometric data, representative plots are shown. The error bars show the mean values with SEMs for at least three independent experiments. *P* values were calculated via two-tailed unpaired *t* tests with Welch's correction; *, *p* < 0.05; **, *p* < 0.01; ***, *p* < 0.001

flow cytometry (Fig. 4c). Both CAR-T cells and Ab- $\gamma\delta$ TCR-T cells exhibited proliferative activity in vitro comparable to that of unmodified $\alpha\beta$ -T cells (Supplementary Data Fig. 11). We measured the cytotoxicity of CAR-T cells and Ab- $\gamma\delta$ TCR-T cells against different tumor cell lines via RTCA. hMSH2-Ab- $\gamma\delta$ TCR-T cells exhibited stronger cytotoxicity against various tumor cell lines with much higher levels of ectopic surface hMSH2 expression, including SW480, NCI-H446, U87-MG, GLC-82, Caco2 and HT-29 cells (Fig. 4d, Supplementary Data Fig. 12a, Supplementary Data Fig. 13a). NCL-Ab- $\gamma\delta$ TCR-T cells also exhibited stronger cytotoxicity against HepG2, OVCAR-8, and MDA-MB-231 cells (Fig. 4e, Supplementary Data Fig. 12b), which presented much higher ectopic NCL expression (Supplementary Data Fig. 13b). Furthermore, hMSH2-Ab- $\gamma\delta$ TCR-T cells exhibited significantly greater cytotoxicity against NCI-H520 cells overexpressing hMSH2 (Supplementary Data Fig. 14a, b). Similar results were observed via LDH cytotoxicity assays for the cytotoxicity of NCL-Ab- $\gamma\delta$ TCR-T cells against NCL-overexpressing A549 cells (Supplementary Data Fig. 14c, d).

To further confirm the antitumor efficacy of hMSH2/NCL-targeted CAR-T cells and Ab- $\gamma\delta$ TCR-T cells in vivo, we established tumor models in NOD. CB17-Prkdcscidll2rgtm1/Bcgen (B-NDG) mice generated via the subcutaneous injection of SW480 cells or HepG2 cells. hMSH2-targeted CAR-T cells and Ab- $\gamma\delta$ TCR-T cells significantly inhibited the growth of tumors (Fig. 4f) and prolonged the survival of tumor-bearing mice (Fig. 4g). Similarly, NCL-CAR-T cells and NCL-Ab- $\gamma\delta$ TCR-T cells significantly inhibited tumor growth in a HepG2 tumor-bearing mouse model, while NCL-CAR-T cells showed slightly better efficacy than NCL-Ab- $\gamma\delta$ TCR-T cells in vivo (Fig. 4h). These results suggest that, similar to hMSH2, NCL can be ectopically expressed on the surface of tumor cells and could serve as a target for tumor immunotherapy.

Ectopic TGM1, a ligand for the V δ 2 TCR

Similar to our verification of NCL as an ectopically expressed ligand of V δ 1 TCRs, we also verified that TGM1 is a novel ectopically expressed ligand of the V δ 2 TCR.

TGM1 is a transglutaminase 1 enzyme that catalyzes the addition of an alkyl group from alkylamine residues to glutamine residues, forming alkylglutamine residues in proteins [27]. We found that menadione triggered ectopic TGM1 expression in normal 293 T cells in a dose (Fig. 5a)- and time (Fig. 5b)-dependent manner. Unpenetrated 293 T cells presented significantly upregulated cell membrane expression of TGM1 after menadione treatment, as shown by confocal microscopy (Fig. 5c), and ectopic expression of TGM1 significantly increased the cytotoxicity of V δ 2 $\gamma\delta$ T cells to 293 T cells (Fig. 5d).

We also examined TGM1 expression in tumor tissues and tumor cell lines and found that ectopic expression of TGM1 was upregulated in some tumor tissues (Fig. 5e) and some tumor cell

lines (Fig. 5f, g). Consistent with the results in menadione-treated 293 T cells, menadione treatment resulted in a significant increase in the ectopic expression of TGM1 on the surface of tumor cells and significantly increased the cytotoxicity of $\gamma\delta$ T cells to tumor cell lines (Fig. 5h, i).

To further verify whether TGM1 is a ligand of $\gamma\delta$ T cells, we demonstrated the specificity of the binding of OT3 and TGM1 via an MST assay (Fig. 5j). Moreover, we amplified human $\gamma\delta$ T cells isolated from PBMCs in vitro. The results showed that the recombinant TGM1 protein could specifically upregulate the expression of the activation markers CD69 and CD25 on V δ 2⁺ $\gamma\delta$ T cells (Fig. 5k–n). Taken together, these findings confirmed that TGM1 could be ectopically expressed on the surface of tumor cells and cells after menadione treatment and serve as a ligand for V δ 2 $\gamma\delta$ T-cell activation and cytotoxicity.

Ligand-induced $\gamma\delta$ T-cell cytotoxicity

Although $\gamma\delta$ T cells are believed to recognize a wide range of structurally different ligands, very few true protein ligands have been identified as being recognized by $\gamma\delta$ TCRs. Functionally, without negative selection during thymus development, $\gamma\delta$ T cells exploit their limited diversity to recognize stress-induced self-protein antigens, including MHC class I chain-related gene A/B (MICA/B) [7, 22], UL-16-binding proteins (ULBPs) [21] and heat shock proteins (HSPs) [23], as well as hMSH2. The finding that $\gamma\delta$ T cells may recognize a series of stress-induced ectopic protein ligands prompted us to explore the importance of these ectopic proteins in the $\gamma\delta$ T-cell-mediated antitumor immune response. We conducted a systematic analysis of the effects of the known stress-induced ligands described above on different tumor cell lines and $\gamma\delta$ T-cell cytotoxicity in vitro.

The results of the cytotoxicity assay revealed that $\gamma\delta$ T cells exhibited significant cytotoxicity against 71.2% of 50 human cell lines without tumor type specificity (Fig. 6a, Supplementary Data Table 4). Correlation analysis revealed a significant positive correlation between the cytotoxicity of $\gamma\delta$ T cells and the binding activity of tumor cells to GTM and OT3 peptides (Fig. 6b), as well as with the ectopic expression of NCL (Fig. 6c).

Flow cytometry was used to assess the expression of all 11 stress-inducible ligands that are recognized by $\gamma\delta$ T cells on 13 different human tumor cell lines (Fig. 6d), and correlation analysis of the expression of these ligands and $\gamma\delta$ T-cell cytotoxicity was performed. The results revealed that the expression of ULBP1, ULBP2 and HSP90 was positively correlated with $\gamma\delta$ T-cell cytotoxicity (Fig. 6e), whereas that of the other genes was not (Supplementary Data Fig. 15). Furthermore, we analyzed the expression levels of stress-induced ligands on the surface of four pairs of tumor cell lines derived from the same tissue type; these cell lines included HeLa and CASK-1 cells (cervical carcinoma), Raji and Daudi cells (B lymphoma), LoVo and HR8348 cells (colorectal

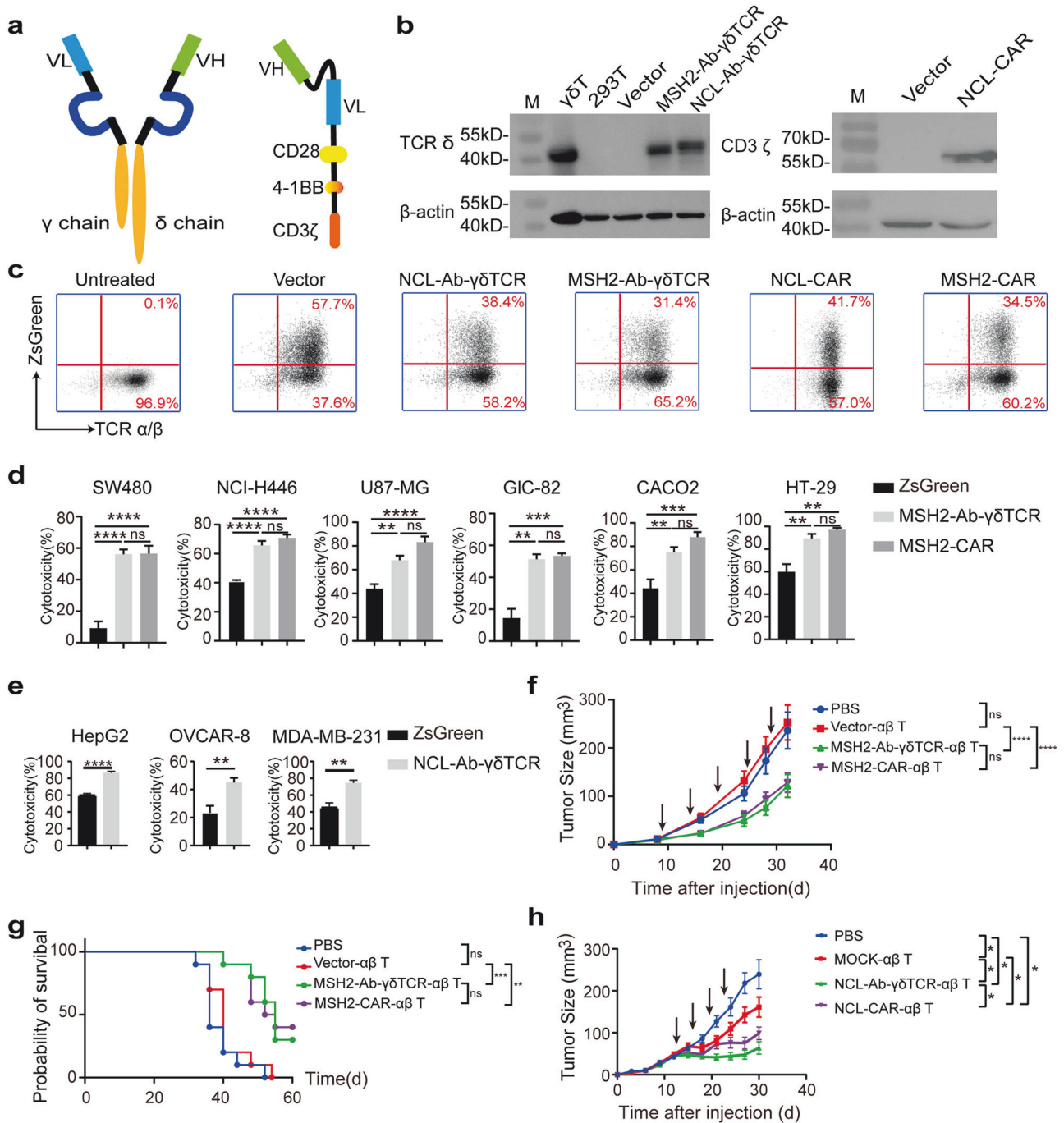
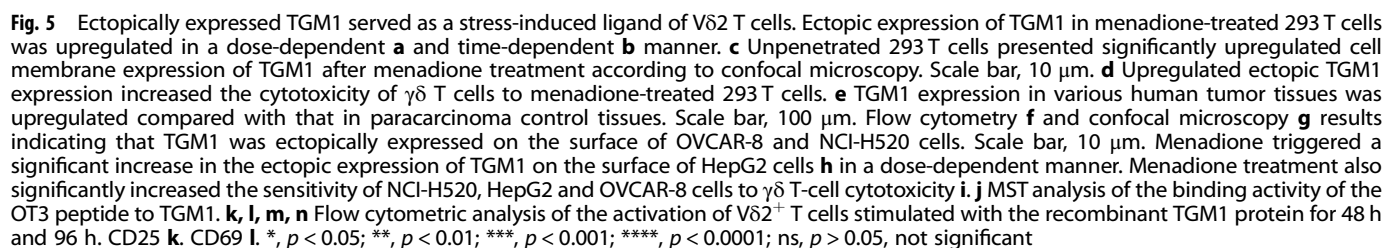


Fig. 4 Construction and functional verification of Ab- $\gamma\delta$ TCR/CAR-T cells that target ectopically expressed hMSH2 and NCL. **a** Schematic of Ab- $\gamma\delta$ TCR (left) and CAR (right) molecules. **b** Western blotting was performed to evaluate the expression of Ab- $\gamma\delta$ TCRs and CARs in 293 T cells with anti-TCR δ chain and anti-CD3 ζ antibodies. **c** Representative flow cytometric plots of the ZsGreen signal showing the transduction efficacy. RTCA was performed to measure the cytotoxicity of hMSH2 Ab- $\gamma\delta$ TCR/CAR-T cells **d** and NCL-Ab- $\gamma\delta$ TCR-T cells **e** against tumor cells in vitro. The plot shows the statistical data at 24 h. The data are representative of three independent experiments and are expressed as the means $\pm$ SDs. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$; ns, $p > 0.05$, not significant. hMSH2-Ab- $\gamma\delta$ TCR-T cells and hMSH2-CAR-T cells exhibited strong cytotoxic activity against tumor cells in the B-NDG mouse model. B-NDG mice implanted with SW480 cells were treated with Vector- $\alpha\beta$ T cells, hMSH2-Ab- $\gamma\delta$ TCR-T cells, or hMSH2-CAR-T cells. Tumor growth in every mouse at each time point was measured with a Vernier caliper. The tumor growth curve **f** and survival curve **g** of each group are shown. The arrow shows the time point for treatment. **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$; ns, $p > 0.05$, not significant. **h** NCL-Ab- $\gamma\delta$ TCR-T cells exhibited strong cytotoxic activity against tumor cells in the B-NDG mouse model. B-NDG mice implanted with HepG2 cells were treated with MOCK-T cells, NCL-Ab- $\gamma\delta$ TCR-T cells or NCL-CAR-T cells. Tumor growth in every mouse at each time point was measured with a Vernier caliper. The arrow shows the time point for treatment

carcinoma), and SMMC-7721 and HepG2 cells (hepatocellular carcinoma), and these cell types were significantly different in their sensitivities to $\gamma\delta$ T-cell cytotoxicity. Compared with those in the $\gamma\delta$ T-cell-sensitive cell lines, the expression levels of most

stress-inducible proteins in the $\gamma\delta$ T-cell-insensitive cell lines were significantly lower (Supplementary Data Fig. 16), confirming that the stress-induced ectopic expression of proteins was closely associated with the cytotoxicity of human $\gamma\delta$ T cells to tumor cells.



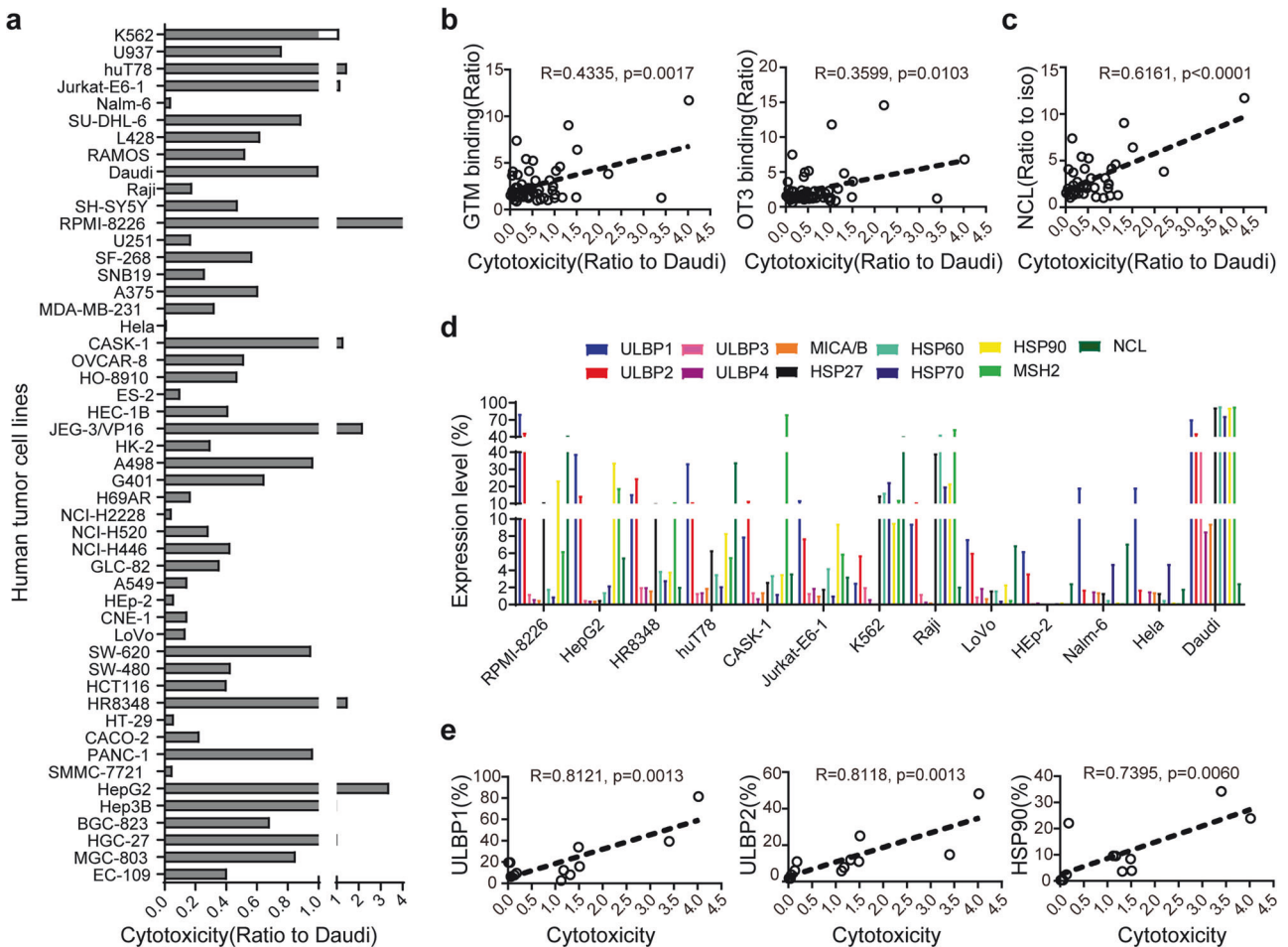


Fig. 6 The cytotoxicity of human $\gamma\delta$ T cells against tumor cell lines was positively correlated with the ectopic expression of $\gamma\delta$ T-cell ligands on tumor cells. A systematic cytotoxicity assay of human $\gamma\delta$ T-cell responses to 50 human tumor cell lines was performed via an LDH assay (E:T = 10:1). **a** The cytotoxicity of $\gamma\delta$ T cells against different target tumor cells is shown as ratios and normalized to their cytotoxicity against Daudi cells. Correlation analysis was performed with the 50 human tumor cell lines to analyze the correlation between their binding activities to GTM/OT3 **b** or their ectopic expression of NCL **c** with their sensitivity to $\gamma\delta$ T-cell cytotoxicity. Flow cytometry was used to detect the expression of 11 stress-inducible ligands that are recognized by $\gamma\delta$ T cells in 13 different human tumor cell lines, and the percentage of positive cells (compared with that of the iso cells) is displayed to indicate the expression level of certain stress-inducible ligands **d**. The expression of 3 of these ligands, ULBP1, ULBP2 and HSP90, was positively correlated with $\gamma\delta$ T-cell cytotoxicity **e**

Unregulated ectopic proteins under radiation treatment

Radiotherapy is one of the major cancer treatment strategies and induces cellular stress through DNA damage, subcellular organelle damage, and reactive oxygen species generation [28]. We applied a radiation cell model to simulate radiotherapy-induced cell stress to observe the changes in ectopic protein expression and subsequently the changes in $\gamma\delta$ T-cell cytotoxicity against target cells. The results revealed that after irradiation, tumor cells exhibited obvious morphological changes (Fig. 7a), and the expression levels of the stress-inducible proteins ULBP1, ULBP2, HSP60 and MICA/B were also significantly increased in irradiated tumor cells (Fig. 7b, Supplementary Data Fig. 17); these effects were accompanied by the significantly increased cytotoxicity of human $\gamma\delta$ T cells against these irradiated cells in vitro (Fig. 7c, Supplementary Data Fig. 18).

Next, we investigated whether low-dose radiation can increase the antitumor effect of $\gamma\delta$ T cells in a transplanted tumor mouse model (Fig. 7d). We found that two ionizing radiation treatments (2 Gy each) at local tumor sites 24 h before the first two $\gamma\delta$ T-cell injections significantly increased the sensitivity of tumors to $\gamma\delta$ T-cell-mediated antitumor activity. Compared with $\gamma\delta$ T cells or radiation treatment alone, combined therapy with $\gamma\delta$ T cells and radiation significantly prolonged the survival of tumor-bearing

mice and inhibited the growth of tumors (Fig. 7e–g). Overall, our results demonstrate that radiation increases the ectopic expression levels of stress-induced protein ligands on the surface of tumor cells and subsequently promotes the antitumor efficacy of $\gamma\delta$ T cells.

DISCUSSION

In the mid-1980s, $\gamma\delta$ T cells were discovered in both mice and humans [29]. During intrathymic development, $\gamma\delta$ T cells emerge before $\alpha\beta$ T cells at the early stages, and in vivo, most mature $\gamma\delta$ T cells home toward mucosal and epithelial tissues, in which $\gamma\delta$ T cells play unique roles in first-line responses to infection and tumor immune surveillance via direct cytotoxicity and the secretion of cytokines, such as IFN- γ and IL-17 [1, 8, 30]. Unlike mouse $\gamma\delta$ T cells, human $\gamma\delta$ T cells are classified into two major subsets according to their δ chain usage. V δ 1 T cells pair with V δ 1 chains to form different V γ chains, which are preferentially enriched in epithelial tissues, whereas V γ 9V δ 2 T cells, which are not present in mice, are distributed in peripheral lymphoid tissue and account for only 1–5% of circulating T lymphocytes [1, 2, 31]. V δ 1 and V δ 2 T cells are almost completely composed of human $\gamma\delta$ T cells, except for a very small number of V δ 3 subsets in the liver.

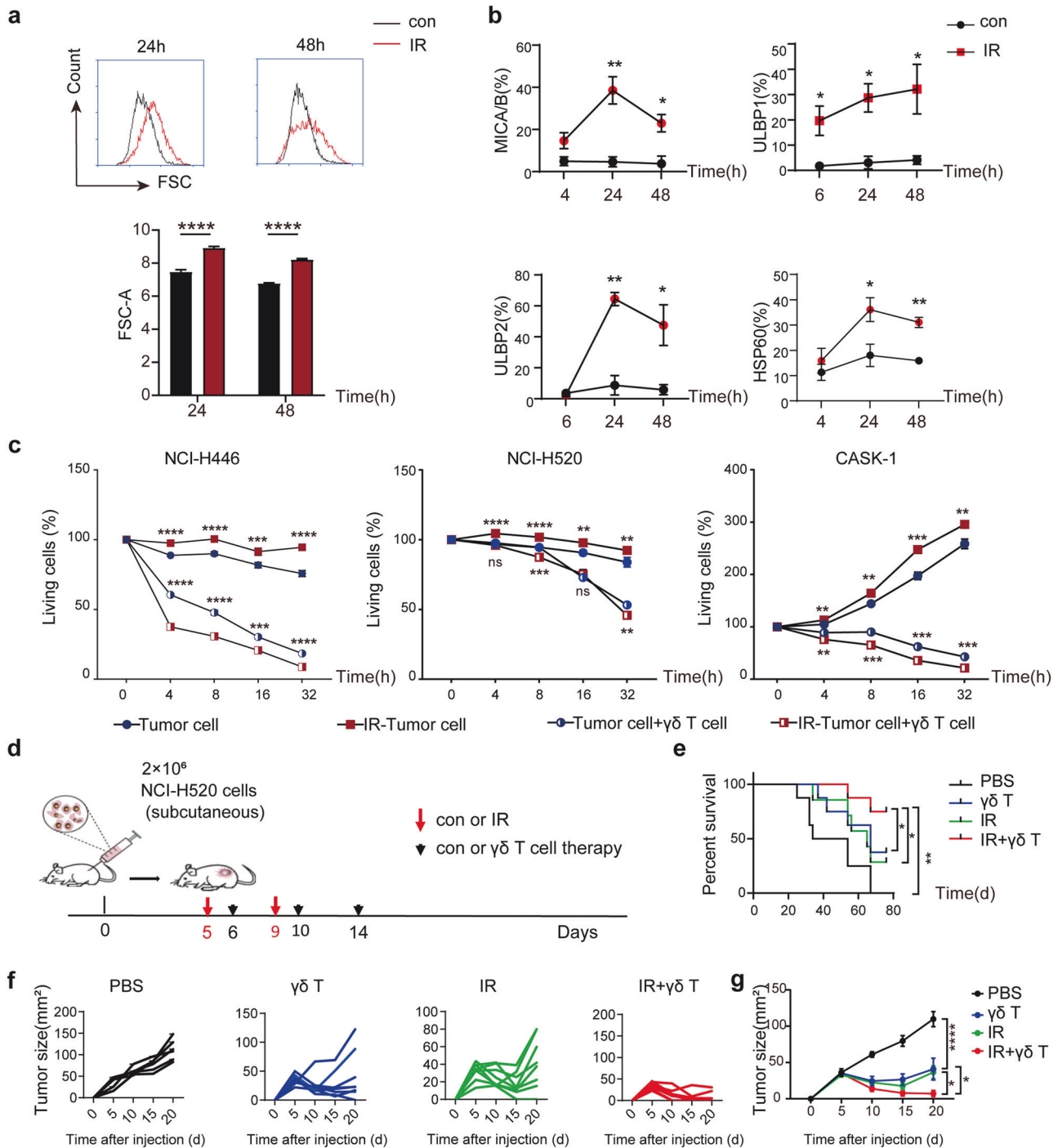


Fig. 7 The combination of $\gamma\delta$ T-cell therapy with radiation increased the antitumor efficacy of human $\gamma\delta$ T-cell immunotherapy by increasing the ectopic expression of stress-induced ligands on irradiated tumor cells. **a** Irradiation exposure can lead to morphological changes in tumor cells. After exposure to 5 Gy radiation, the FSC of NCI-H520 cells significantly increased from 24 h to 48 h, indicating that the cells were in a state of stress. **b** Irradiation upregulated the ectopic expression of the ligands ULBP1, ULBP2, HSP60 and MICA/B on NCI-H520 cells in vitro. Flow cytometric results showing ligand expression on the tumor cell membrane before and at different time points after irradiation. **c** The cytotoxicity of human $\gamma\delta$ T cells to irradiated tumor cells was significantly increased. The cytotoxicity of $\gamma\delta$ T cells to three kinds of tumor cells treated with or without 5 Gy ionizing radiation was recorded via RTCA. **d** Design of the in vivo experiment in mouse transplanted tumor models. $\gamma\delta$ T-cell adoptive immunotherapy alone or in combination with radiotherapy was compared in this study. The combination of immunotherapy with low-dose radiation therapy prolonged survival **e** and promoted tumor regression in tumor-bearing mice. The tumor growth of each mouse in each group is shown **f**, and the average value of all the groups is shown **g**. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$ (ANOVA). For survival curves, ANOVA and the log-rank (Mantel–Cox) test were performed. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$

Without CD4 or CD8 molecules, $\gamma\delta$ T cells do not seem to undergo both positive and negative selection during thymus development. The principal antigenic specificity of an immunoglobulin derives from its most diverse CDR3 region of the heavy chain. Owing to the similarity of $\gamma\delta$ TCRs and immunoglobulins in direct ligand binding [1], CDR3 δ serves as the key determinant for the specificity of $\gamma\delta$ TCR antigen binding, with minimal assistance by the γ chain. The critical role of CDR3 δ in antigen recognition by $\gamma\delta$ T cells has been well confirmed [15]. Owing to the limited diversity of the $\gamma\delta$ TCR repertoire [32], $\gamma\delta$ TCRs do not generally bind to protein antigens such as peptides presented by MHC molecules in a one-to-one manner [33, 34] but tend to recognize some nonpeptidic antigens, self-MHC-like molecules and stress-induced self-proteins directly. However, the specific recognition pattern of $\gamma\delta$ T cells for these protein antigens remains unclear.

One of the core roles of immune recognition is to distinguish between “self” and “nonself”. Innate immune cells, such as epithelial or myeloid cells, respond to foreign and internal environments via the interaction of pattern recognition receptors (PRRs) with PAMPs or DAMPs, and NK cells respond to the abnormal absence of self-MHC I or increased stress-related self-ligands on somatic cells. B cells bind to antigens directly through diverse B-cell receptors (BCRs), and $\alpha\beta$ TCRs recognize antigenic peptides and MHC complexes on antigen-presenting cells, both of which are antigen specific. Through structural approaches to examine the roles of $\gamma\delta$ TCRs in antigen recognition, we found that $\gamma\delta$ TCRs primarily use conserved flanking regions of the CDR3 δ sequence to bind ligands [18], suggesting a limited number of $\gamma\delta$ TCR ligands.

On the basis of tumor-specific CDR3 δ OT3, we previously validated ectopically expressed hMSH2 as a protein ligand of the V δ 2 TCR [21] and hypothesized that stress-induced membrane-expressing proteins serve as “danger signals” to trigger $\gamma\delta$ T-cell-mediated responses. In this study, we used two types of probes, a GTM-grafted V γ 4 δ 1-Fc fusion protein and a biotin-labeled V δ 2 CDR3 OT3 peptide, which basically cover the two main kinds of human TCR cell populations, to screen for additional protein ligands of $\gamma\delta$ TCRs. Among the 21,000 proteins, we identified 16 intracellular candidates and validated ectopically expressed NCL and TGM-1 as protein ligands of the V δ 1 and V δ 2 TCRs, respectively. Furthermore, we found that these stress-inducible ligands could be upregulated on the surface of tumor cells after radiation exposure and could then be recognized by $\gamma\delta$ T cells or several ligand-targeting engineered CAR-T cells and TCR-T cells, subsequently mediating the “recognition as killing” process to eliminate tumor cells.

A previous report demonstrated that only phosphorylated NCL can interact with the peptide-binding domain of heat shock cognate 70 (HSC70), which is a chaperone, resulting in the ectopic expression of NCL from the intracellular region to the cell surface [35]. NCL undergoes multiple posttranslational modifications, such as phosphorylation, glycosylation, and acetylation, in vivo [25, 36]. The interaction between the recombinant NCL protein and GTM is weak in MST, but immunoprecipitation assays and functional experiments have shown that NCL can stimulate the expansion, activation, and cytotoxic activity of V δ 1 T cells, which indicates that the need for molecular chaperones and posttranslational modifications is highly important for the interaction of cell surface-expressed NCL with V δ 1 T cells.

Answering the fundamental immunological question of how $\gamma\delta$ T cells recognize protein antigens is highly important. That is, both the V δ 1 and V δ 2 subsets of human $\gamma\delta$ T cells can recognize certain ectopically expressed self-proteins on the cell surface, including NCL and hMSH2, which are produced because of exposure to stress, such as infection or malignant transformation, and play roles as first-line defense mediators and sensors for immunosurveillance.

First, unlike NK cell activation, which involves mutual regulation of activating receptors and inhibitory receptors in a concise and rapid manner via long-term evolution, according to the tissue distribution of $\gamma\delta$ T cells, stress-induced ectopic expression of single or multiple proteins leads to a rapid local response of $\gamma\delta$ T cells, direct cytotoxicity, or the secretion of cytokines for inflammation and the regulation of subsequent adaptive immune responses.

In contrast to the $\alpha\beta$ TCR, the conserved flanking region of the CDR3 δ sequence plays a crucial role in antigen recognition, which determines the innate property in which the protein ligands recognized by the $\gamma\delta$ TCR are limited. The unique antigen recognition pattern of $\gamma\delta$ TCRs to certain ectopically expressed self-proteins can also be considered as a special version of pattern recognition and a supplement to the pattern recognition mechanism.

Previous studies on the recognition of nonpeptide antigens by $\gamma\delta$ T cells are limited; however, proteins are the most advanced form of living substances and exhibit greater diversity in their changes and more universal importance in their roles. The recognition of stress-induced protein expression by $\gamma\delta$ T cells occurs under a range of different circumstances, including tumors, cellular aging, noninfectious inflammation, infection, intracellular symbiosis, autoimmune diseases, and immune homeostasis imbalance, indicating broader application prospects for $\gamma\delta$ T-cell-based immunotherapy.

Taken together, our results indicate that $\gamma\delta$ T cells, which have limited TCR diversity, can perform innate-like functions and recognize ectopically expressed intracellular self-proteins on many different cell types as DAMPs under stress conditions. This finding highlights a pioneering antigen recognition mechanism of both V δ 1 and V δ 2 TCRs to protein antigens. Moreover, immunotherapeutic strategies, for example, combination with radiotherapy, which synergistically promotes the potency of $\gamma\delta$ T-cell therapy by increasing the expression of stress-induced ectopic proteins, or Ab-TCR-T-cell or CAR-T-cell therapy, which directly targets stress-induced ectopic proteins, can provide new ideas for the treatment of tumors, as well as infection, aging and autoimmune diseases.

MATERIALS AND METHODS

Clinical tumor samples

Liver puncture biopsy samples were obtained from 6 hepatocellular carcinoma patients who participated in an investigator-initiated trial (NCT04032392). The trial was approved by the Ethics Committee of the Fifth Medical Center of Chinese People's Liberation Army Hospital in Beijing, China. The main information corresponding to the hepatocellular carcinoma samples is provided in Supplementary Data Table 1. The samples were soaked in 4% paraformaldehyde (PFA) tissue fixative solution for 24 h before being embedded in paraffin and sectioned for experiments.

Cell lines

All the human tumor cell lines used are listed in Supplementary Data Table 2. The cell lines used in this study were obtained from the Cell Resource Center, Peking Union Medical College, which is part of the National Science and Technology Infrastructure, the National Biomedical Cell-Line Resource, and stored in our laboratory.

Tissue immunohistochemistry

To analyze the binding of OT3 or GTM peptides to human tumor tissues and the expression of NCL and protein-glutamine γ -glutamyltransferase K (TGM1) in different human tumor tissues, a tissue microarray (Biomax) containing 12 kinds of cancer tissues and paracarcinoma tissues was used. After deparaffinization, the tissue microarray was washed with phosphate-buffered saline (PBS) and incubated in 3% H₂O₂ for 15 min to inactivate endogenous peroxidase. Then, antigen retrieval and tissue permeabilization were performed with 0.25% Triton X-100. The nonspecific sites of the tissue microarray were blocked with normal goat serum, and then, the microarray was immunostained with an appropriate concentration of

biotin-conjugated OT3 or GTM peptide and incubated at 4 °C overnight. After washing with 1× TBST, an HRP-conjugated streptavidin antibody was added, and the mixture was incubated at room temperature (RT) for 1 h.

For the NCL/TGM1 expression assay, mouse anti-human NCL antibody C23 (Santa Cruz Biotechnology, 1:500)/rabbit anti-human TGM1 antibody (AbClonal, 1:150) and HRP-conjugated anti-mouse IgG/HRP-conjugated anti-rabbit IgG (ZSGB-Bio) were added and incubated. The tissue microarray was washed, and 3'-diaminobenzidine (DAB) was added and incubated. Then, hematoxylin was used to rest the tissue, and the microarray was sealed with resin.

For analysis of the expression of NCL and 11 other protein ligands recognized by $\gamma\delta$ T cells in liver puncture biopsy samples, paraffin sections of the samples were heated at 60 °C for 1 h, cooled naturally and then dewaxed. After being washed in PBS, the sections were heated in sodium citrate antigen repair solution, thus allowing the exposure of antigens. Eleven of the antibodies used in the assay were obtained from Santa Cruz Biotechnology [anti-HSP27, anti-HSP60, anti-HSP70, anti-HSP90, anti-ULBP1, anti-ULBP2, anti-ULBP3, anti-ULBP4, anti-MSH2, anti-NCL and anti-MICA/B antibodies (mouse)], and the rabbit anti-AGR2 antibody was obtained from Abcam. These antibodies were diluted in goat serum at a ratio of 1:250, added dropwise to the tissue and incubated overnight at 4 °C. Horseradish peroxidase (HRP)-conjugated anti-mouse IgG/HRP-conjugated anti-rabbit IgG (ZSGB-Bio) was used as described above for subsequent detection.

Flow cytometric analysis

1. For analysis of the binding of the OT3 or GTM peptide to different human tumor cells, the tumor cells were washed and resuspended in FACS buffer (PBS with 1% BSA). Single-cell suspensions were incubated with 20 μ g of biotin-conjugated OT3 or GTM peptide at 4 °C for 1 h. Then, the cells were stained for 30 min at 4 °C with an APC-conjugated streptavidin antibody (BioLegend).
2. For analysis of NCL expression in tumor cell lines, 1 μ g of rabbit anti-human NCL (GeneTex) primary antibody was added, and the mixture was incubated for 30 min at RT. The secondary antibody, Alexa Fluor 647-conjugated goat anti-rabbit IgG (CST), was diluted to 1:500 and incubated for 10 min at RT. For analysis of hMSH2 expression in tumor cell lines, 1 μ g of mouse anti-human MSH2 (Abcam) primary antibody was added, and the mixture was incubated for 1 h at 4 °C. FITC-conjugated goat anti-mouse IgG (BioLegend) at a 1:100 dilution was added, and the mixture was incubated for 30 min at 4 °C. For analysis of TGM1 expression, TGM1 polyclonal rabbit antibodies (AbClonal) were used as primary antibodies at a 1:100 dilution and incubated for 1 h. FITC-conjugated anti-rabbit IgG was then added at a 1:150 dilution and incubated for 30 min on ice. For the cell stress model, 293 T cells and tumor cells were treated with menadione (MedChemExpress) at certain concentrations and for certain durations. Menadione (20 μ M) was used in the flow cytometric assay to assess the expression of NCL and TGM1. Tumor cells were exposed to 50 μ M menadione for 15 min to simulate short-term, high-intensity stress.
3. To determine the total percentages of $\gamma\delta$ T cells, CD69⁺ $\gamma\delta$ T cells, CD25⁺ $\gamma\delta$ T cells, V δ 1 T cells and V δ 2 T cells from amplified PBMCs, 1 μ g of FITC-conjugated mouse anti-human $\gamma\delta$ TCR (BioLegend), PE-conjugated mouse anti-human CD69 (BioLegend), APC-conjugated mouse anti-human CD25 (BioLegend), FITC-conjugated mouse anti-human V δ 1 TCR (Thermo Fisher) and PE-conjugated mouse anti-human V δ 2 TCR (BioLegend) were added, and the mixture was incubated for 30 min at 4 °C.
4. For the sorting assay, peripheral blood mononuclear cells (PBMCs) were 2-color stained with PE/Cy7-conjugated anti-CD3 and biotin-conjugated anti-human TCR $\gamma\delta$ (BioLegend), with APC-conjugated streptavidin (BioLegend) as a secondary antibody. After stimulation for 2 and 4 days, the cells were subjected to 5-color staining with PE/Cy7-conjugated anti-CD3 (BioLegend), APC-conjugated anti-V δ 1 TCR (Miltenyi), APC/Cy7-conjugated anti-CD69 (BioLegend), Brilliant Violet 510-conjugated anti-TCR V δ 2 (BioLegend), and Percp/Cy5.5-conjugated anti-human CD25 (BioLegend) antibodies.
5. For analysis of stress-induced ligand expression, different tumor cells or cells treated with ionizing radiation at a dose of 5 Gy for 6, 24 or 48 h were collected for flow cytometric analysis. The cells were collected and counted, and 1×10^6 cells in each group were

incubated with 1% BSA and then stained with a primary antibody at 4 °C for half an hour. After washing, the cells were stained with a secondary antibody at 4 °C for half an hour and assessed by flow cytometry. Primary antibodies, including mouse anti-human ULBP3, ULBP4, MICA/B, HSP27, HSP60, HSP70, HSP90, hMSH2, and NCL; rabbit anti-human ULBP1 and ULBP2; and isotype mouse IgG1 and IgG2a antibodies, were obtained from Santa Cruz. FITC-conjugated anti-rabbit and FITC-conjugated anti-mouse secondary antibodies were purchased from BioLegend.

6. To evaluate the proliferation of T cells after lentiviral transduction, peripheral blood mononuclear cells (PBMCs) from healthy donors were harvested, adjusted to a concentration of 1×10^6 cells/mL and subsequently stained with 1 μ M CellTrace™ Far Red per mL of cell suspension at 37 °C for 20 min in the dark. T-cell culture medium (RPMI-1640 with 10% FBS, 2 mM glutamine and 200 U/mL recombinant human IL-2) was added, and the mixture was incubated for 5 min. All the cells were harvested and resuspended in cold T-cell culture medium, and 1×10^6 cells were seeded into well plates and activated with T-Cell TransAct (Miltenyi). After 24 h, activated T cells were infected with lentivirus at a multiplicity of infection (MOI) of 2. The proliferative capacity of Ab- $\gamma\delta$ TCR- or chimeric antigen receptor (CAR)-T cells was determined by flow cytometry at 48 h after the removal of lentiviral stimulation.

Except for the sorting analysis, all flow cytometric analyses were performed with a BD Accuri C6 system, and all flow data were analyzed via C6 software. The sorted cell suspensions were analyzed via a FACS Cannto II system (Becton Dickinson). The acquired data were analyzed via Flow Jo software (Becton Dickinson).

Microscale thermophoresis (MST)

The OT3 or GTM peptides were labeled with 1 μ L of FITC-streptavidin (BioLegend) for 1.5 h on ice, and the fluorescence-labeled peptides were diluted to a concentration that yielded a fluorescence count between 200 and 1500. Next, stress-inducible proteins, including HSP27/60/70/90, hMSH2, MICA/B, and ULBP1/2/4/6 (Sino Biological, Inc.), were diluted to an appropriate concentration with buffer solution. Then, the protein and fluorescence-labeled peptide were mixed at a ratio of 1:1 for 5 min at RT, the samples were loaded into the capillaries, and the MST experiment was run with 40% MST power.

Protein microarray

The microarray slides (CD) were initially blocked with 3% BSA-TBST at RT for 1 h. Serum was added, and the samples were incubated with the arrays for 1 h at RT. In the GTM-Fc protein microarray, GTM-Fc or Fc (Sino Biological, Inc.), as the primary antibody, was diluted, added, and the mixture was incubated at RT for 1 h. Then, a Cy3-conjugated goat anti-human IgG antibody (Bioss) was added, and the mixture was incubated for 1 h at RT. In the GTM- $\gamma\delta$ TCR microarray, soluble Vy4 δ 1 TCR with the GTM sequence in its CDR3 δ sequence was used, followed by an anti-pan $\gamma\delta$ TCR antibody and a Cy3-labeled goat anti-mouse IgG antibody (Bioss). A control experiment was carried out with a recombinant soluble Vy9 δ 2 TCR. In the OT3 protein microarray, OT3 peptide or biotin, as the primary antibody, was diluted, added, and incubated at 4 °C for 12 h. Cy3-conjugated streptavidin (Bioss) was used to probe the primary antibodies bound to the slide at RT for 1 h.

For all the assays, a rabbit anti-GST monoclonal antibody (CST) was added and incubated at RT for 1 h. An Alexa Fluor 647-conjugated goat anti-rabbit IgG (CST) secondary antibody was added and incubated at RT for 1 h. Finally, the microarray slides were scanned with a GenePix 4000B fluorescent microarray scanner (GBI). The probe signals were acquired via GenePix Pro 6.0 software (Molecular Devices).

Immunoprecipitation

The full-length NCL sequence was obtained from the American Type Tissue Culture Collection (ATCC) and cloned and inserted into CFUGW plasmids via Phusion DNA polymerase (Biolabs). The experimental procedures were performed according to the manufacturer's instructions. The NCL plasmids were subsequently analyzed by sequencing. A vector with the correct sequence was transfected into 293 T cells via Lipofectamine 2000 (Invitrogen). Twenty microliters of anti-GFP-conjugated Sepharose beads (Biotool) were added to 293 T whole-cell lysates and incubated for 2 h at 4 °C with rotation. GTM-Fc (2 μ g) and Fc

(2 µg) were added to the system and incubated overnight at 4 °C with rotation. The pellets were washed six times with cell lysis buffer. Finally, the pellets were resuspended in 20 µL of 1× SDS sample buffer and heated to 95 ~ 100 °C for 2 ~ 5 min to remove the Sepharose beads. The samples were analyzed by Western blotting using an anti-human IgG antibody (ZSGB-Bio).

Biolayer interferometry (BLI) assay

The BLI assay was performed with a ForteBio Octet Red 96 instrument to determine the affinity of GTM-TCR for FBXO2. In brief, 6×His-tagged GTM-TCR was immobilized onto Ni-NTA-coated biosensors and then dipped into a solution containing FBXO2 protein for 6 min, followed by dissociation for 5 min. Sensorgrams of the concentration series were corrected with corresponding blank curves and fitted globally with Octet evaluation software via a 1:1 Langmuir model of binding. The affinity constant (KD) (mol/L) reflects the binding capacity.

Confocal microscopy

1. Tumor cells, including HT-29 and BGC-823 cells, were placed on slides pretreated with polylysine overnight and fixed with 4% cold PFA. Fixed cells were permeabilized or not permeabilized and incubated with rabbit anti-human NCL (GeneTex) overnight at 4 °C. Alexa Fluor 647-conjugated goat anti-rabbit IgG antibody (CST) was then added, and the samples were incubated for 2 h at RT. Rabbit IgG (GeneTex) was used as an isotype control. The slides were examined via confocal microscopy (Zeiss).
2. For analysis of TGM1 expression, NCI-H520 cells or menadione-treated 293 T cells (for 15 min) and rabbit anti-human TGM1 (ABclonal, 1:100) and Alexa 488-conjugated anti-rabbit (A11008, 1:1000) antibodies were used following the same protocol.

Menadione-induced cell stress model

NCI-H520 cells were treated with 0.5, 5, 10, 20, 40, or 80 µM menadione (MedChemExpress; untreated and DMSO-treated cells were used as controls) for 15 min and then collected for analysis of total protein expression. Western blotting with an anti-PAR monoclonal antibody (ENZO Life Sciences, 1:2000) was used to detect the characteristic modifications of PAR. Anti-β-actin (Sigma, 1:5000) was used for normalization. A Cell Counting Kit-8 assay (Dojindo Laboratories) was used to evaluate the effects of different treatment times and concentrations of menadione on the viability of both 293 T cells and the tumor cell line NCI-H520. The assay was conducted according to the manufacturer's instructions.

γδ T-cell expansion with recombinant proteins

Freshly isolated PBMCs from healthy donors were cultured in RPMI 1640 medium (Gibco BRL) supplemented with 10% FCS and IL-2 (200 U/mL) in 24-well culture plates supplemented with immobilized recombinant NCL (CUSABIO, 20 µg/mL). Plates with immobilized BSA (Sigma, 20 µg/mL) were used as a control. After 1 week of culture, the purity and subtype of the γδ T cells were analyzed via flow cytometry.

γδ T cells were sorted from freshly isolated PBMCs from healthy donors and cultured in RPMI 1640 medium (Gibco BRL) supplemented with 10% FBS and IL-2 (40 U/mL, Quanqi) in 96-well culture plates with immobilized recombinant NCL (CUSABIO, 20 µg/mL) and TGM1 (IPODIX, 20 µg/mL). The plates with immobilized medium, BSA (Sigma, 20 µg/mL), anti-Vδ1 TCR (Beckman Coulter, 1 µg/mL) and zoledronic acid (Kelun, 1 µmol/L) were used as controls. After 48 and 96 h of culture, the percentages of CD69⁺ and CD25⁺ Vδ1 T and Vδ2 T cells were analyzed via flow cytometry.

Sorting γδ T cells from PBMCs

PBMCs were separated from peripheral blood via density gradient centrifugation via Ficoll-Hypaque density gradient centrifugation fluid (MD Pacific). After cell counting and viability detection, the PBMCs were successively incubated with an anti-human TCRγ/δ antibody conjugated with biotin (BioLegend) and antibiotic GMP microbeads (Miltenyi, 170-076-709). Finally, γδ T cells were magnetically separated.

Expansion of Vδ1 T cells from PBMCs

PBMCs were isolated from healthy donors via a Ficoll gradient prior to cryopreservation. Then, the PBMCs were thawed and plated onto an immobilized agonistic anti-Vδ1 TCR antibody for 2–4 days at a concentration of 1×10^6 cells/mL. Human cytokines (100 ng/mL IL4, 70 ng/mL IFN-γ, 7 ng/mL IL-21, and 15 ng/mL IL-1β) were added to the medium in the first stage for 8 days. The cells were incubated at 37 °C and 5% CO₂. In the second stage, fresh medium supplemented with cytokines (70 ng/mL IL-15 and 30 ng/mL IFN-γ) was replaced every 5 to 6 days until day 21. The percentage of Vδ1 TCRs was analyzed via flow cytometry.

Enzyme-linked immunosorbent assay (ELISA)

Vδ1 T cells (purity over 90%) that were expanded from PBMCs with an anti-Vδ1 TCR monoclonal antibody (Immunotech) were cultured in medium without IL-2 for 24 h before stimulation with NCL (CUSABIO, 50 µg/mL) or BSA (Sigma, 50 µg/mL) for 36 h and PMA/ION for 4 h. Human IFN-γ and IL-2 ELISA kits (4 A Biotech) were used to measure the concentrations of IFN-γ and IL-2 in the cell supernatants. The experimental procedure followed the manufacturer's instructions.

Lactate dehydrogenase (LDH) cytotoxicity assay

1. A systematic cytotoxicity assay was performed to assess the effects of human γδ T cells on 50 human tumor cell lines. Different tumor cell lines (5×10^4 per well) were plated in round-bottom plates (96 wells). Human γδ T cells expanded from healthy PBMCs by incubation with immobilized anti-γδ TCR antibodies for 10 days (purity > 90%) were added at an effector:target (E:T) ratio of 10:1 for 6 h. All the groups were assayed in triplicate. The cytotoxicities of the γδ T cells against different target tumor cells are shown relative to the cytotoxicities of the γδ T cells against Daudi cells to avoid differences in the cytotoxicity of different effector γδ T cells, which were expanded from different healthy donors.
2. To assess Vδ1 T-cell cytotoxicity against tumor cells, HepG2, MDA-MB-231, and OVCAR-8 cells were added to 96-well plates at a density of 3×10^4 cells/well and incubated with anti-NCL or isotype antibodies for 6 h at 37 °C. Vδ1 T cells were added to the plates as effector cells at an E:T ratio of 10:1, and each condition was assayed in triplicate.
3. To analyze the cytotoxicity of γδ T cells to menadione-treated tumor cells, HepG2, NCI-H520, and OVCAR-8 cells were treated with 50 µM menadione for 15 min (untreated cells were used as controls) and then used as target cells. γδ T cells (E:T = 5:1) were added and incubated for 6 h at 37 °C.
4. For radiation induction, Daudi cells were exposed to ionizing radiation at a dose of 5 Gy, with untreated cells serving as controls. The irradiated Daudi cells were then cocultured with γδ T cells (1×10^5) for a cytotoxicity assay at an effector-to-target (E:T) ratio of 10:1 in a final volume of 100 µL per well at various time points following radiation exposure.

We assessed cytotoxicity via an LDH assay via the CytoTox 96 Nonradioactive Cytotoxicity Assay Reagent Kit (Promega). The cytotoxicity assay was conducted according to the manufacturer's instructions.

Real-time cellular analysis (RTCA)

The medium of the tumor cell lines was adjusted to the proper concentrations, and 1×10^4 cells were seeded into well plates and monitored via RTCA (xCELLigence S16, ACEC Biosciences). The tumor cells were cultured for 24 h to allow cell adherence before the effector cells were added. The number of adjacent tumor cells (living target cells) was continuously recorded. Cytotoxicity was quantified by subtracting the percentage of viable target cells cultured alone from that of target cells cocultured with effector cells.

For radiation induction, adjacent tumor cells were treated with ionizing radiation at a dose of 5 Gy (untreated cells were used as controls). γδ T cells (1×10^5) were added at an E:T ratio of 10:1 in a final volume of 100 µL/well. Tumor cells were cultured alone or cocultured with γδ T cells for another 32 h.

For the in vitro cytotoxicity assay, MOCK-αβ T, hMSH2-Ab-γδ TCR-αβ T, NCL-Ab-γδ TCR-αβ T or hMSH2-CAR-αβ T cells (5×10^4 ZsGreen⁺ cells) were added at an E:T ratio of 5:1 in a final volume of 200 µL/well. Tumor cells

were cultured alone or cocultured with different effector cells for another 24 h.

γδ CAR-T-cell construction

The variable region of the TCR γ 9 and TCR δ 2 chains with the tumor antigen-specific CDR3 region OT3, which was sequenced from γδ T cells infiltrating ovarian epithelial carcinoma tissues, was directly synthesized by the TSINGKE Company. The sequences were subsequently cloned and inserted into an MSCV retro vector containing the sequence CD8-CD28BBZ. The amplified γδ2-CD8-CD28BBZ sequence was then cloned and inserted into a cFUGW lentiviral vector containing a Flag tag at the N-terminus and GFP. The γδ CAR vector was transfected into 293 T cells in a 10-cm dish with the psPAX2 packaging plasmid and PMD2. G envelope plasmid following the instructions for jetPRIME transfection reagent (Polyplus). Then, the 0.45-μm filtered virus supernatant was concentrated by ultrafiltration and resuspended in RPMI-1640 medium.

PBMCs from healthy donors were isolated via density gradient centrifugation and activated with 1 μg/mL human CD3 NALE HIT3a and 5 μg/mL human CD28 NALE CD28.2 antibodies (BD) for 24 h. T cells were then transduced with γδ CAR lentivirus at an MOI of 2 and centrifuged at 1800 rpm for 60 min at 30 °C. After 20–24 h, the activated antibodies and lentivirus were removed via centrifugation, the T cells were cultured for another 48 h in RPMI-1640 medium supplemented with 10% FBS, 2 mM glutamine and 200 U/mL recombinant human IL-2, and the GFP signal corresponding to the γδ CAR-positive T cells was evaluated via flow cytometry. All the results were obtained from samples from no fewer than 3 donors.

Generation of Ab γδ TCR-T cells and CAR-T cells targeting hMSH2 and NCL

The sequences of the variable regions of the anti-hMSH2 and anti-NCL antibodies were derived from CN 110218251 A and WO 2017/156032 A1, respectively. The sequences of the conserved regions of the human TCR γ and δ chains were obtained from IMGT. The CAR is a third-generation CAR containing the CD28/41BB/CD3ζ domains. After sequence synthesis, Ab-γδ TCRs and CARs were cloned and inserted into the pLVX lentiviral vector via homologous recombination for delivery into T cells. PBMCs from healthy donors were isolated via density gradient centrifugation and activated with T-Cell TransAct (Miltenyi). After 24 h, activated T cells were infected with lentivirus at an MOI of 2 and cultured in RPMI-1640 medium supplemented with 10% FBS, 2 mM glutamine and 200 U/mL recombinant human IL-2. The transduction efficacy of Ab-γδ TCRs and CARs in T cells was determined by flow cytometry. All results with the Ab-γδ TCR and CAR were obtained with samples from no fewer than 3 donors.

Western blotting

For colorectal carcinoma cell lines, cells were collected for analysis of FBXO2 expression via anti-FBXO2 antibodies (Proteintech).

After 24 h of plasmid transfection, the 293 T cells were collected for analysis of total TCR δ (Santa Cruz), CD3Zeta (Abcam), and β-actin (Sigma) protein expression.

For T cells, after 4 days of expansion, lentivirus-infected T cells were collected for analysis of total protein expression. Antibodies recognizing CD3Zeta (Abcam), Flag (Sigma), β-actin (Applygen) and GFP (Chemotek) were added and incubated overnight at 4 °C. All the primary antibodies were detected via HRP-conjugated anti-mouse or anti-rabbit antibodies (ZSGB-Bio), which were added and incubated at RT for 2 h. A chemiluminescence solution (Thermo Fisher) was used to detect the expression of the proteins.

Verification of Ab-γδ TCR-αβ T-cell specificity

The NCI-H520 and A549 cell lines were transfected with hMSH2-GFP and NCL-GFP (Addgene) plasmids, respectively. After 24 h, the expression of GFP was observed via fluorescence microscopy. A total of 1×10^4 tumor cells (NCI-H520/NCI-H520-hMSH2) were seeded into well plates, and hMSH2-Ab-γδ TCR-αβ T cells were added to achieve an E:T ratio of 5:1 in a final volume of 200 μL/well. The results were monitored by RTCA (xCELLigence S16, ACEC Biosciences) for 24 h as described above. For NCL-Ab-γδ TCR-αβ T cells, A549/A549-mock/A549-NCL cells were used as target cells, and untreated αβ T, MOCK-αβ T, or NCL-Ab-γδ TCR-αβ T cells were employed as effector cells at an E:T ratio of 5:1. We evaluated cytotoxicity via an LDH assay, as described above.

Animal model and treatment

Animal experiments were performed according to the guidelines of the Animal Use Committee of the Experimental Animal Center, Institute of Basic Medical Sciences, Chinese Academy of Medical Sciences.

For radiotherapy, 2×10^6 NCI-H520 lung cancer tumor cells were subcutaneously engrafted into the backs of BALB/c nude mice, and 32 mice were randomly divided into four groups when the tumor diameter reached 5–9 mm. The mice were then given ionizing radiation at a dose of 2 Gy at the local tumor site once every four days (two total treatments) and injected with 1×10^7 γδ T cells near the tumors 24 h after radiation.

For γδ CAR-T cells, OVCAR-8 cells were transfected with luciferase-IRES-GFP viral supernatant, and the resulting cell clones were sorted via flow cytometry on the basis of the expression of GFP. Female 6–8-week-old NCG mice were obtained from the Model Animal Research Center of Nanjing University. Stable OVCAR-8 cells labeled with luciferase were subcutaneously injected (3×10^6 cells per mouse) in the right flank. On days 7, 10, 14, and 18 after tumor injection, GFP-αβ T cells or γδ CAR-αβ T cells (1×10^7 cells per mouse) were intratumorally injected. Then, luciferase expression was detected by using D-Luciferin Firefly in vivo Imaging Reagents (PerkinElmer), and the results were analyzed via live imaging software. The animal experiments were performed according to the guidelines of the Animal Use Committee of the Experimental Animal Center, Institute of Basic Medical Sciences, Chinese Academy of Medical Sciences.

For experiments with Ab-γδ TCR-T and CAR-T cells, female 6–8-week-old B-NDG mice (10 for each group) were obtained from Biocytogen Pharmaceuticals (Beijing). SW480 cells and HepG2 cells were obtained from the Cell Center of the Chinese Academy of Medical Sciences and injected subcutaneously (5×10^6 cells per mouse) into the right flank. For the SW480 tumor-bearing mouse model, MOCK-αβ T or hMSH2-Ab-γδ TCR-αβ T cells (2×10^6 cells per mouse) were intratumorally injected 4 times on days 9, 13, 17, 21 and 25 (with PBS as a control), and hMSH2-CAR-αβ T cells (1×10^7 cells per mouse) were injected only once, on day 9. For the HepG2 tumor-bearing mouse model, MOCK-αβ T or NCL-Ab-γδ TCR-αβ T cells (1×10^6 cells per mouse) were intratumorally injected 4 times on days 13, 16, 19 and 22 (with PBS as a control), and NCL-CAR-αβ T cells (5×10^6 cells per mouse) were intratumorally injected only once, on day 13 after tumor injection. Tumors were measured with a Vernier caliper every 3 or 4 days, and the volume was calculated according to the following equation: tumor volume (mm^3) = maximal length (mm) $\times$ perpendicular width² (mm^2)/2. When the tumor volume exceeded 2000 mm^3 , the mice were sacrificed, and the survival time was recorded.

Statistical analyses

Statistical analyses of flow cytometric data, γδ T-cell expansion data and in vitro cytotoxicity assay data were performed via appropriate statistical tests, including paired or unpaired two-tailed *t* tests with Welch's correction as needed and one-way analysis of variance (ANOVA) with multiple comparisons (SPSS 13.0 software).

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AUTHOR CONTRIBUTIONS

H.C., J.Z. and H.W. conceived the project. H.Y. performed the NCL screening and function verification. X.Z. and C.L. generated the ectopic protein-targeting Ab TCR-T cells and CAR-T cells. M.C. performed the TGM1-related experiments. Y.P. and D.T. performed the $\gamma\delta$ CAR-T-cell experiments. J.H. and Y.Z. conducted the radiation-related experiments. Q.D., M.C. and S.W. performed the screening and cytotoxicity of $\gamma\delta$ T cells to different tumor cell lines. Y.X., Y.H. and P.D. developed the methodology. H.C. and Y.Z. performed the data analysis and wrote the manuscript. J.Z. and H.W. modified the manuscript.

COMPETING INTERESTS

No potential conflicts of interest were reported by the authors. H.W. is an editorial board member of Cellular & Molecular Immunology, but he has not been involved in the peer review or the decision-making of the article.

ADDITIONAL INFORMATION

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